

LAB EXPERIMENT RECORD**01-10-2026--Set 4: Product Inventory Management**

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Aim

To visualize product inventory data using a bar chart, stacked bar chart and scatter plot, and to build an interactive dashboard for exploring the inventory data.

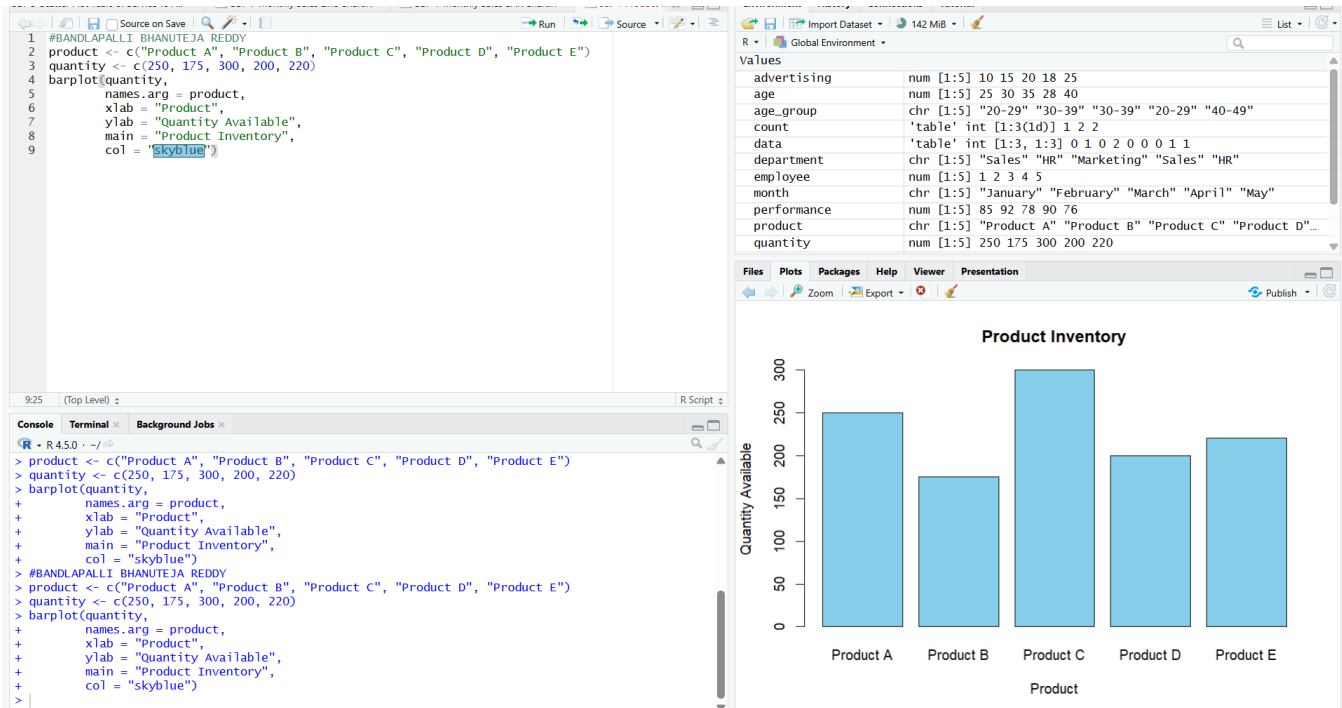
Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Question 1: Bar Chart of Product Quantity

Question: Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Aim: To visualize the quantity available for each product using a bar chart with labelled axes and a title.

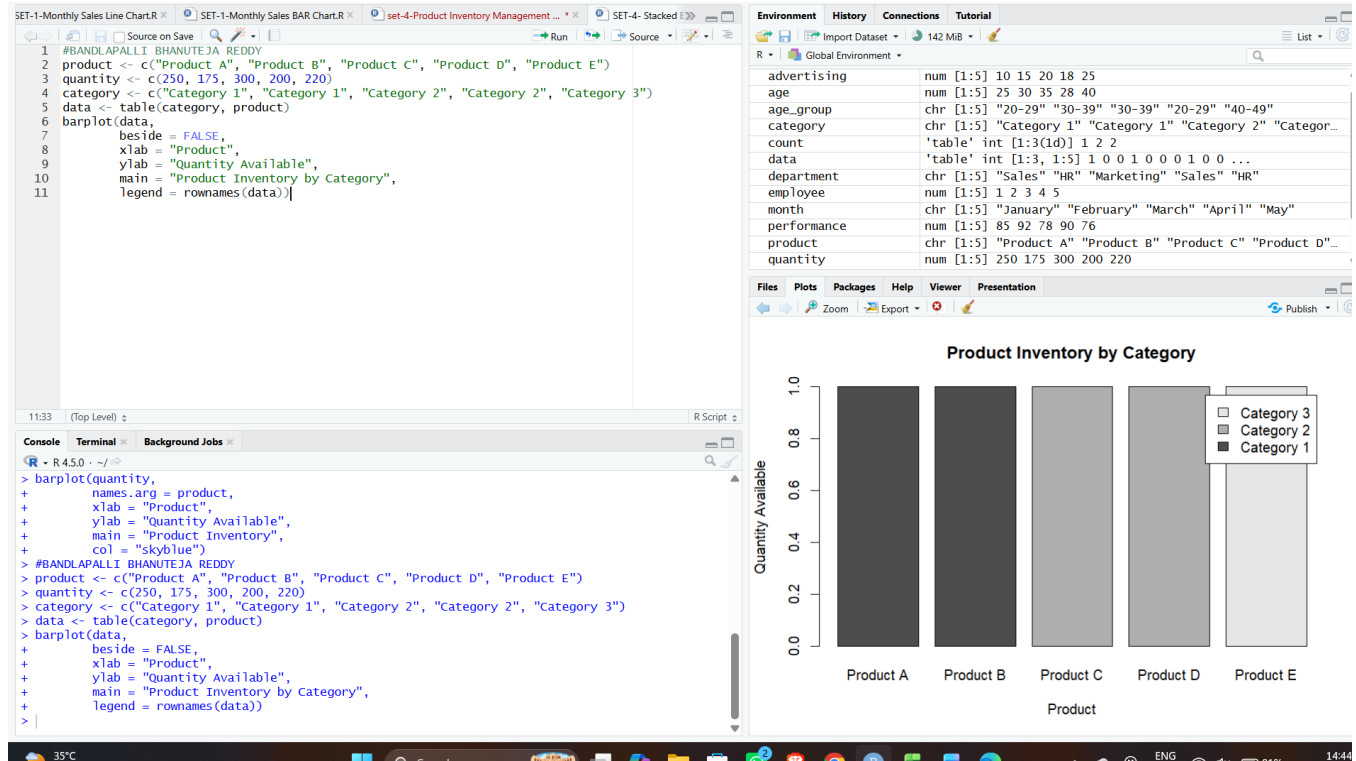


Result: Product C has the highest stock (300) and Product B the lowest (175), so Product B needs restocking first.

Question 2: Stacked Bar Chart by Product Category

Question: Generate a stacked bar chart to show the quantity of each product within different product categories.

Aim: To show how the quantity of each product is divided across product categories using a stacked bar chart.

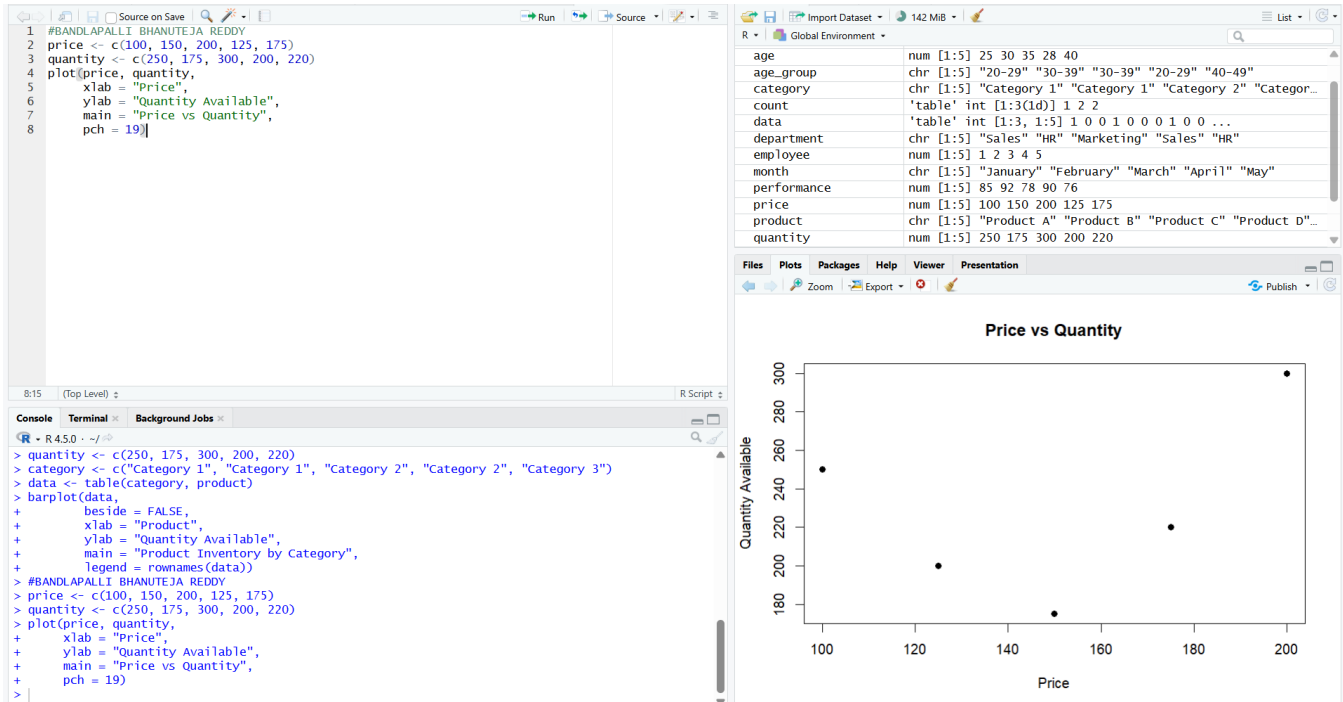


Result: Each bar is split by category, showing each category's share of a product's stock; the dataset gives no category column, so categories must be supplied.

Question 3: Scatter Plot of Price vs Quantity

Question: Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

Aim: To explore the relationship between product price and quantity available using a scatter plot.

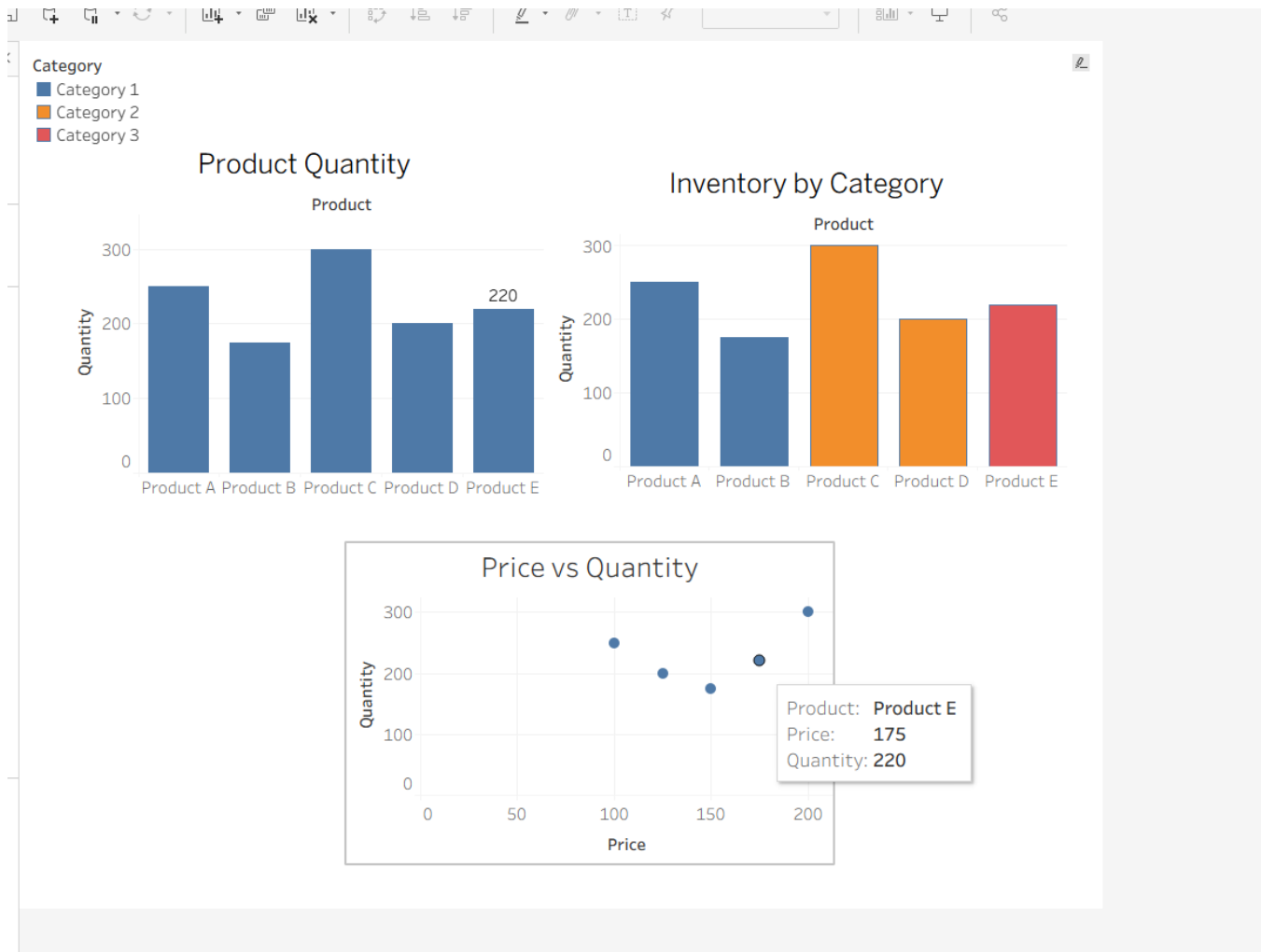


Result: The plot shows whether higher-priced products are stocked in lower quantities; the dataset gives no price column, so a conclusion needs price values.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Aim: To build an interactive Tableau dashboard that combines the bar chart and stacked bar chart for exploring inventory data.



Result: The dashboard places both charts together, and filters let users compare stock across products and categories interactively.

Overall Result

Observation: Product C has the highest quantity (300) and Product B the lowest (175), with 1145 units in total across the five products.